Wearable device and analysis system for early detection and screening of Breast cancer

Over view of the Technology :

Breast Cancer is now the most common cancer in India and accounts for approximately 27% of all cancers in women. Early detection of breast cancer has the potential to achieve 100% cure of the disease. In India, a stigma towards regular screening for cancer exists due to fear and ignorance. Reliance on Western guidelines employing regular mammography is impractical in India due to the large population of young patients with more dense breasts along with the high cost of the machines and trained manpower and staff requirements. Therefore a more practical alternative should be there to identify potential women in a community **by quick initial screening process** and only suspected cases may be referred for further diagnosis. C-MET has developed a wearable device by using high sensitivity thermal sensors for the early detection and screening of breast cancer. The initial clinical trials were conducted in 117 patients and around 200 volunteers and the results are in line with the standard diagnostic tools such as mammogram, ultrasound and CT scan. This device can be operated with minimum training and is portable. This device does not inflict any pain or involve any radiation exposure and above all the privacy of the woman is ensured.

C-MET has filed one US patent (Application **15/926,935,** Date of Filing: **20/03/2018)** and two Indian patent applications (No.: **201741017186 and** No.: **201711047118)** for this technology.

Features and Technical Specifications

The salient features and advantages of the device are

- Privacy of the women is ensured
- Highly trained Man power is not required. Even an ASHA worker can operate
- Portable, Works on battery power. Ideal for rural India
- Economical
- Do not inflict any pain to the women
- Suitable for younger women also
- No radiation exposure
- Early detection of breast cancer- decrease mortality rate

The technology consists of design and methodology of manual fabrication of wearable device for early detection and screening of breast cancer and an analysis system which gives 2D image of breast thermal profile. This consists of Wearable device having different sizes and 2D stand alone software for imaging (executable file only).



The level of technology readiness at C-MET

The technology readiness level is 7

Period of training under the agreement

C-MET, Thrissur will give one month training at the premises of C-MET. Fabrication of 3 numbers of wearable devices will be demonstrated. With each of these 3 wearable devices, Volunteers trial run will be demonstrated for at least one volunteer.

The ToT package contains the following

1. Document containing raw material/component details
2. Processing details
3. Product specification

4. Test plan and procedure

5. Training.

Note:

1. Optionally, for interested parties, Technology on thermal sensor probes (based on one type of chip thermistor) for wearable device will be given on additional ToT fee.

2. Optionally, interested parties, for Technology on data acquisition system may contact C-DAC, Thiruvanathapuram

ToT Cost:

(i) The ToT cost for <u>Wearable device and analysis system for early detection and screening of Breast cancer is Rs. 8.86 Lakhs + Applicable taxes</u>

(ii) The additional ToT cost for Technology on thermal sensor probes (based on one type of chip thermistor) for wearable device is Rs 2.01 Lakhs + Applicable taxes